Week 2 Shell Scripting Machine Test

Question 1: Student Management System

Requirements

• Store 5 student names in an array.

• Display all students.

• Display total student count.

• Print each student using a loop.

• Search for a student name entered by the user.

• Display 'Student Found' or 'Student Not Found'.

Question 2: Employee Information Tool

Requirements

• Accept employee name, department, and salary as command-line arguments.

• Display all entered information.

• Display script name.

• Display total number of arguments.

• Display all arguments passed.

Question 3: Number Analyzer

Requirements

• Accept a number as a command-line argument.

• Check whether it is positive or negative.

• Check whether it is even or odd.

• Check whether it is greater than 100.

Question 4: Course File Manager

Requirements

• Create a file named courses.txt.

• Add Linux, Docker, Kubernetes, and AWS to the file.

• Display file contents.

• Display total number of lines in the file.

• Check whether the file exists.

Question 5: Employee List Reader

Requirements

• Read employee names from a file.

• Display each employee name.

• Display total employees processed.

Question 6: Server Monitoring Simulator

Requirements

• Store server names in an array.

• Loop through all servers.

• Display checking status for each server.

• Display total servers checked.

Question 7: Backup Utility

Requirements

• Accept a folder name as a command-line argument.

• Display backup start message.

• Display backup folder name.

• Display backup completion message.

• Show usage message if no argument is provided.

Question 8: DevOps Inventory Management System

Requirements

• Create a menu-driven shell script.

• Add Server.

• View Servers.

• Search Server.

• Count Servers.

• Exit the application.

Bonus Challenge: Log Report Generator

Requirements

• Read a log file containing ERROR, INFO, and WARNING entries.

• Count each log type.

• Generate a summary report showing totals.